Functions - Trigonometric Functions: Part 1

Course: Pre-Calculus-I

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26CS(Evening) - Fall 2026

Outline

1 Angles and the Unit Circle

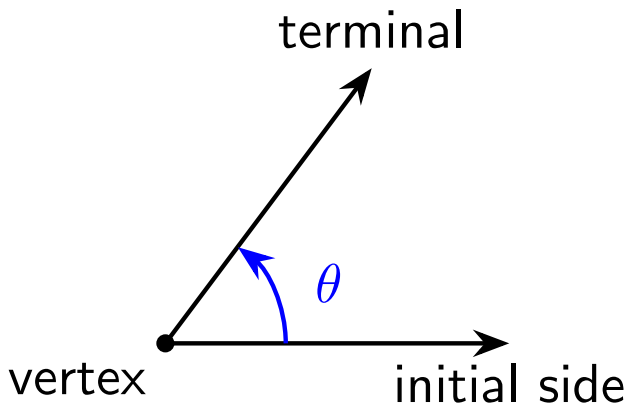
2 The Sine and Cosine Functions

Angles: Degrees and Radians

An angle is formed by rotating a ray from an initial side to a terminal side about a vertex.

Two common measures:

- **Degrees:** full turn $= 360^\circ$.
- **Radians:** full turn $= 2\pi$ rad.



Conversion:

$$\pi \text{ rad} = 180^\circ$$

$$1^\circ = \frac{\pi}{180} \text{ rad}, \qquad 1 \text{ rad} = \frac{180^\circ}{\pi}.$$

Note

In this course we mostly use radians, because they make the trigonometric functions natural functions on $\mathbb{R}$.

Standard Angles

Degrees	0°	30°	45°	60°	90°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$

Examples of conversion:

- $120^\circ = 120 \cdot \dfrac{\pi}{180} = \dfrac{2\pi}{3}$ rad.
- $\dfrac{3\pi}{4}$ rad $= \dfrac{3\pi}{4} \cdot \dfrac{180}{\pi} = 135^\circ$.

Note

Angles are often measured counter-clockwise from the positive x-axis (standard position).

The Unit Circle

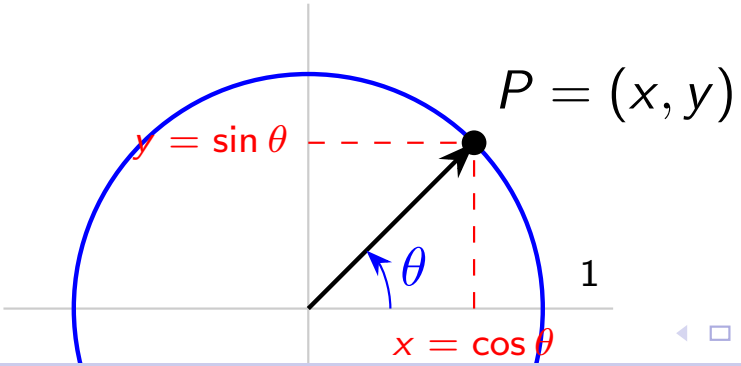
Definition

The unit circle is the circle of radius 1 centred at the origin:

$$x^2 + y^2 = 1.$$

For an angle θ in standard position, let $P = (x, y)$ be the point where the terminal side meets the unit circle. Then:

$$\boxed{\cos\theta = x, \qquad \sin\theta = y}$$



Extending to All Real Angles

The point P can rotate around the circle, so θ can be *any* real number.

- $\theta > 0$: counter-clockwise rotation.
- $\theta < 0$: clockwise rotation.
- Adding 2π brings P back to the same place.

Periodicity

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

Note

Because of this periodic behaviour, sin *and* cos *are defined on all of* $\mathbb{R}$ *— just like the other functions we studied.*

The Sine and Cosine Functions

Definitions

For any real number θ, let $P = (x, y)$ be the point on the unit circle at angle θ. Then

$$\boxed{\cos\theta = x, \qquad \sin\theta = y.}$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Note

$\sin\theta$ *and* $\cos\theta$ *are just the coordinates of the point* P *— no need to memorise a separate "formula".*

Domain and Range of sin and cos

Since $P = (x, y)$ always lies on $x^2 + y^2 = 1$:

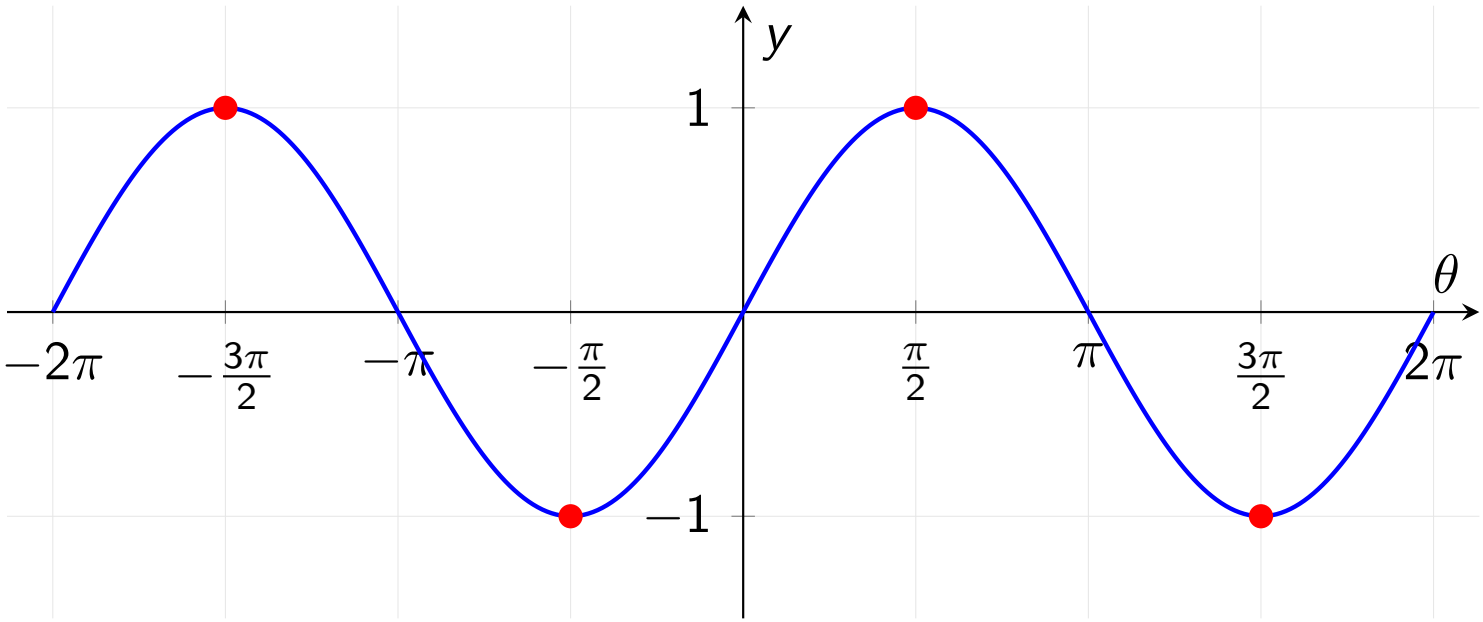
$$-1 \leq x \leq 1, \qquad -1 \leq y \leq 1.$$

Function	Domain	Range
$f(\theta) = \sin\theta$	$\mathbb{R} = (-\infty, \infty)$	$[-1, 1]$
$f(\theta) = \cos\theta$	$\mathbb{R} = (-\infty, \infty)$	$[-1, 1]$

Key consequences:

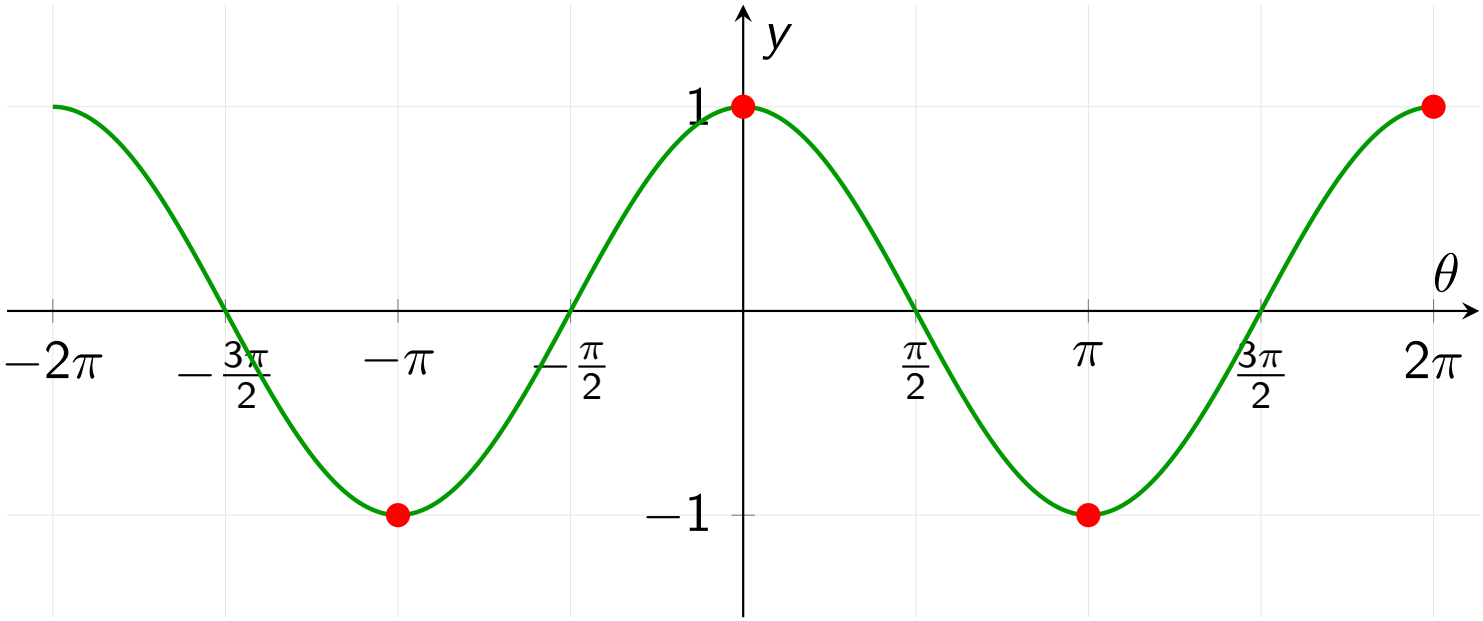
- Both functions oscillate between -1 and 1.
- They achieve the value 1 (max) and -1 (min) *infinitely often*.
- These maximum and minimum values will be used when we study extrema.

Graph of $f(\theta) = \sin\theta$



The sine curve is a smooth wave with **amplitude** 1, **period** 2π, and passes through the origin.

Graph of $f(\theta) = \cos\theta$



The cosine curve is the sine curve shifted left by $\pi/2$; it starts at its maximum $(0, 1)$.

Amplitude, Period, and Phase Shift

For a trigonometric function of the form

$$f(\theta) = A\sin(B\theta + C) + D \quad \text{or} \quad g(\theta) = A\cos(B\theta + C) + D,$$

we have:

Quantity	**Value**
Amplitude	$\lvert A\rvert$
Period	$\dfrac{2\pi}{\lvert B\rvert}$
Phase shift	$-\dfrac{C}{B}$
Vertical shift	D
Range	$[D - \lvert A\rvert,\ D + \lvert A\rvert]$

Note

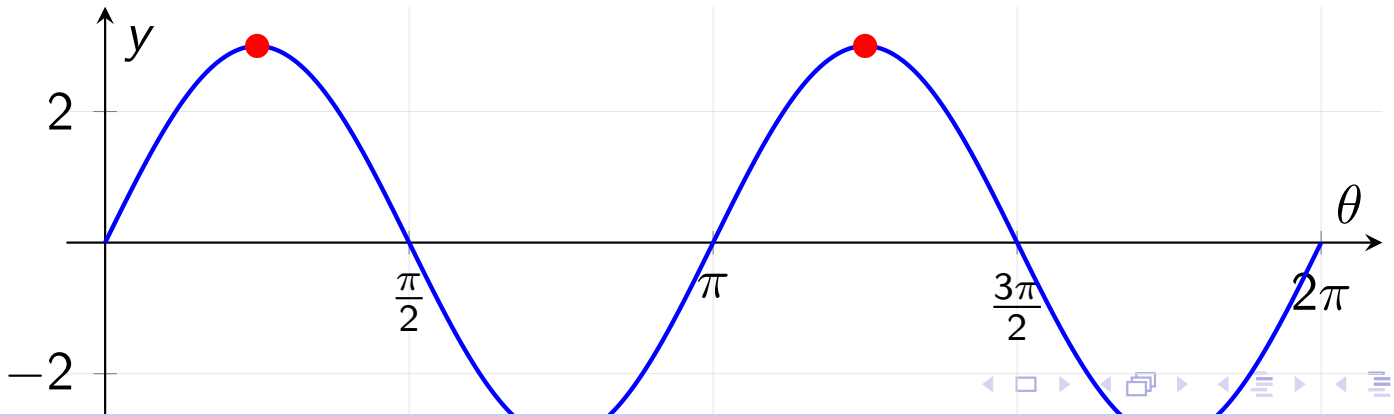
The range follows immediately because $-1 \leq \sin, \cos \leq 1$.

Example 1: $f(\theta) = 3\sin(2\theta)$

Compare with $A\sin(B\theta + C) + D$**:**

$$A = 3, \quad B = 2, \quad C = 0, \quad D = 0.$$

- **Amplitude:** $|A| = 3$.
- **Period:** $\dfrac{2\pi}{2} = \pi$.
- **Range:** $[-3, 3]$.
- Domain: $\mathbb{R}$.
- Passes through $(0, 0)$, reaches max 3 at $\theta = \pi/4$, min -3 at $\theta = 3\pi/4$.



Example 2: $f(\theta) = 2\cos(\theta) + 1$

Here $A = 2,\ B = 1,\ C = 0,\ D = 1$.

- **Amplitude:** 2.
- **Period:** 2π.
- **Vertical shift:** $+1$ (up).
- **Range:** $[1-2,\ 1+2] = [-1,\ 3]$.
- **Maximum** value $= 3$ (achieved when $\cos\theta = 1$).
- **Minimum** value $= -1$ (achieved when $\cos\theta = -1$).

